Constructing lattice surfaces with given Veech groups

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Outline

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1.1: Translation surfaces

1.2: Veech groups

§2: Computing Veech group elements

§3: Constructing lattice surfaces

§1 : Introduction

1.1 – Translation surfaces

Three equivalent definitions

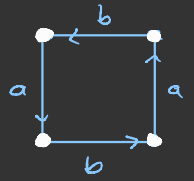
- Defi.: A translation surface is

  (i) a collection of polygons together with edge identifications, such that each edge is identified with one other equal-length, parallel, oppositely-oriented edge (up to 'cut-and-paste' equivalence)

  (ii) a real surface X such that off of a finite set Σ, the subsurface $X \setminus \Sigma$ comes equipped with a translation atlas which extends to give conical singularities at points of Σ of angles $2\pi(d+1)$, $d \in \mathbb{Z}_{\geq 0}$

  (iii) a pair (X, ω), where X is a Riemann surface and ω is a non-zero holomorphic 1-form

Examples



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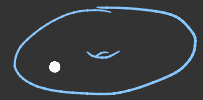

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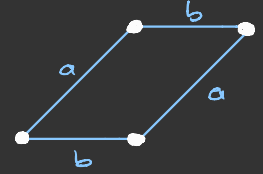

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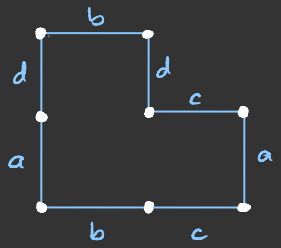

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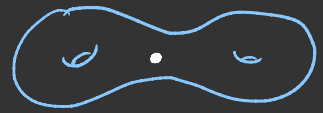

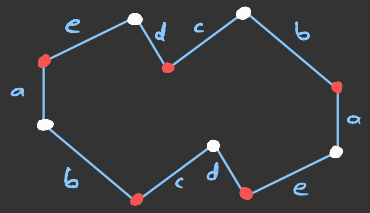

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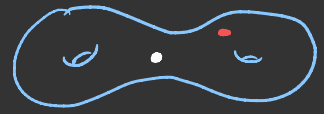
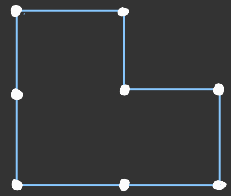


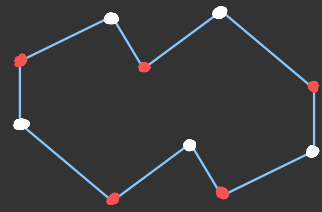
Strata

- Defn.: Denote by $\mathcal{H}(d_1, \dots, d_k)$ the **stratum** of closed (compact, no boundary) and connected translation surfaces with singularities $\Sigma = \{\sigma_i\}_{i=1}^{k}$ of cone angles $2\pi(d_i+1)$, $1 \le i \le k$.

$\mathcal{H}(0)$



$\mathcal{H}(2)$

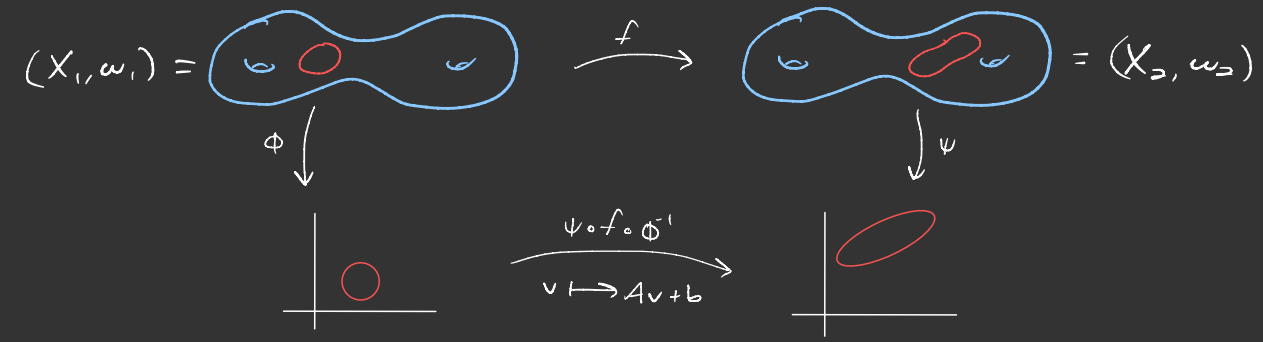


$\mathcal{H}(1,1)$

1.2 - Veech groups

Affine diffeomorphisms

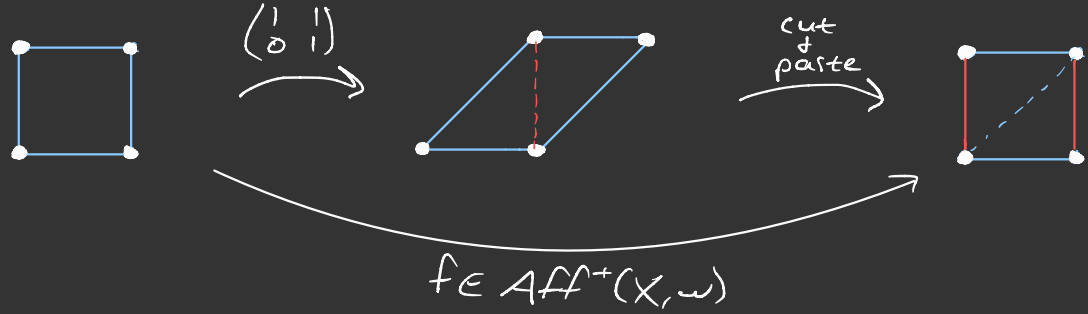
- Defn.: An **affine diffeomorphism** $f : (X_1, \omega_1) \to (X_2, \omega_2)$ is a homeomorphism sending Σ_1 to Σ_2, and such that on $X_1 \setminus \Sigma_1$, f is locally affine with constant Jacobian.



- Defn.: Let $\mathrm{Aff}^+(X, \omega)$ denote the group of all orientation-preserving affine automorphisms of (X, ω).

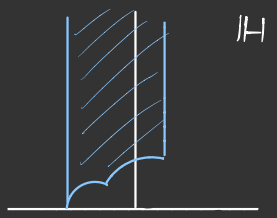
Veech group

- Def'n.: The Veech group of (X,ω), denoted $SL(X,\omega)$, is $der(Aff^+(X,\omega))$, where $der: Aff^+(X,\omega) \to GL_2^+\mathbb{R}$ maps $f \in Aff^+(X,\omega)$ to its Jacobian



Remarks

- For closed, connected (X,ω), $SL(X,\omega) \leq SL_2\mathbb{R}$ is discrete, hence $PSL(X,\omega) := SL(X,\omega)/\langle -Id \rangle$ is a Fuchsian group



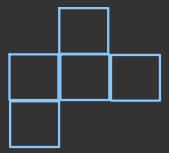
- A classification of which subgroups of $SL_2\mathbb{R}$ are realized as Veech groups is unknown [HMSZ 06]

- Generic (X,ω) have trivial Veech groups. However, (X,ω) with 'especially large' $SL(X,\omega)$ are of particular interest...

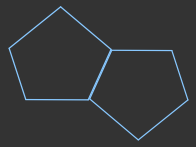
Lattice surfaces

- Defn.: Call (X, ω) a lattice (or Veech) surface if $SL(X,\omega)$ is a lattice (i.e. $SL(X,\omega)$ has finite covolume in $SL_2\mathbb{R}$, or $\nu_{\mathbb{H}}(PSL(X,\omega)\backslash\mathbb{H}) < \infty$)

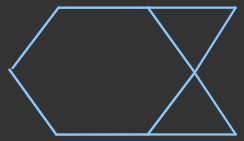
- Ex's:



Square-tiled

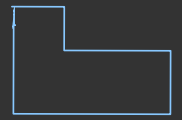


Covers of regular polygons



Ward + Boum-Möller

'Unfoldings' of some triangles



Some L-shapes, ...

§2: Computing Veech group elements

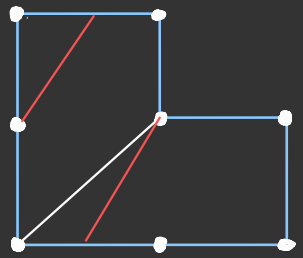
History

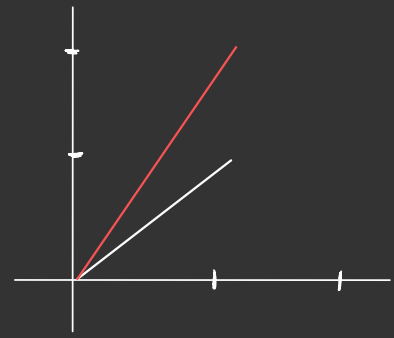
- Algorithms:
  - Square-tiled surfaces: [Sch04]
  - Covers of regular polygons: [Fr08]
  - General case: [Bo10] ([Ve11], [BrJ12]), [Mu17], [Ed17] ([ESS22])

Saddle connections & holonomies

- **Defn.**: A saddle connection is a geodesic γ on (X,ω) with endpoints in Σ and interior in $X \setminus \Sigma$. Its holonomy vector is $\int_\gamma \omega \in \mathbb{C} \approx \mathbb{R}^2$.

  Set $\mathrm{hol}(X,\omega) := \{ \int_\gamma \omega \mid \gamma \text{ a s.c. on } (X,\omega) \}$





'Naive' approach to $SL(X,\omega)$ - membership

- Recall $f \in Aff^+(X,\omega)$ sends Σ to Σ, so $der(f) \in SL(X,\omega)$ satisfies $der(f) \cdot hol(X,\omega) = hol(X,\omega)$
- Does $SL(X,\omega) = \{ A \in SL_2\mathbb{R} \mid A \cdot hol(X,\omega) = hol(X,\omega) \}$?

$$hol\left(\square\right) = hol\left(\square\right) = \{ (x,y) \in \mathbb{Z}^2 \mid gcd(x,y) = 1 \}$$

$$SL\left(\square\right) = SL_2\mathbb{Z} \neq SL\left(\square\right)$$

- Idea: Create a more discerning analogue of holonomy vectors

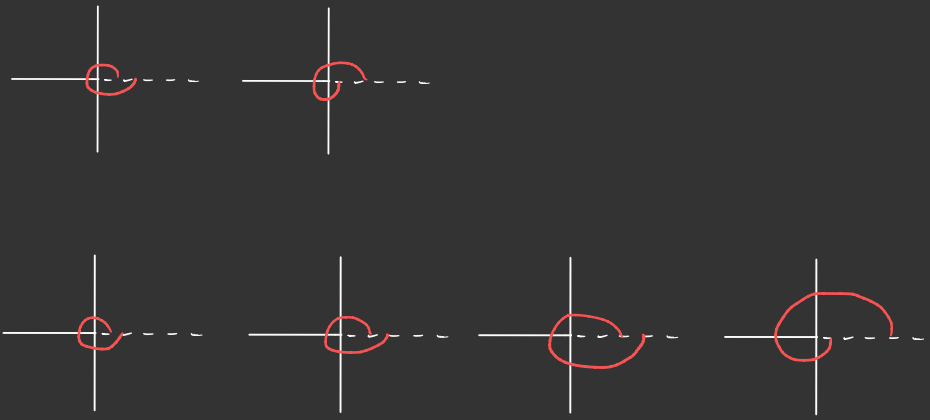
A canonical surface

- Defn.: Fix $H(d_1, \dots, d_k)$. For each $1 \le i \le k$, set $\Theta(d_i) := (\mathbb{C}, z^{d_i} dz)$, and let

$$\Theta = \Theta(d_1, \dots, d_k) := \bigsqcup_{i=1}^{k} \Theta(d_i)$$

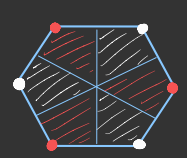
- Ex.: $\Theta(1,3)$

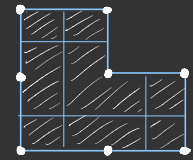
$$\Theta(1,3) = \Theta(1) \sqcup \Theta(3)$$



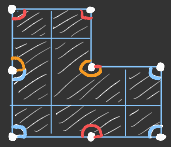
Representing (X,ω) in $\mathcal{O}$

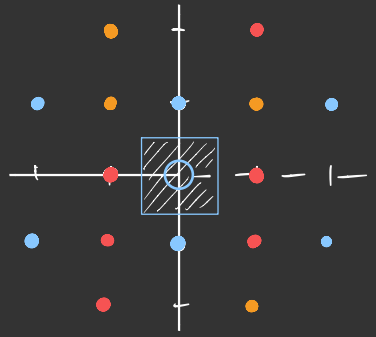
- Every (X,ω) has a (unique) Voronoi decomposition subordinate to Σ

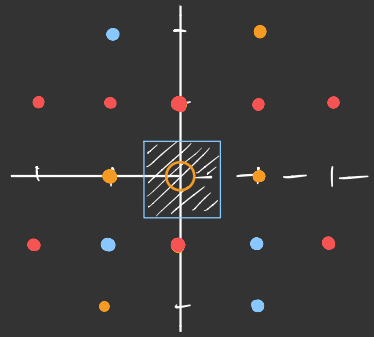


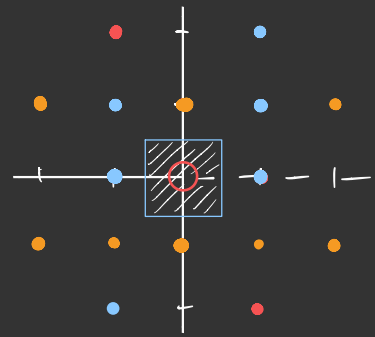


- Use generalized polar coordinates about each $\sigma \in \Sigma$ to develop Voronoi cells and saddle connections in $\mathcal{O}$.









- Defn.: Denote by $\hat{\mathcal{U}}(X,\omega)$ the set of all orientation-paired developments of saddle connections in $\mathcal{O}$, called o.p. marked segments.

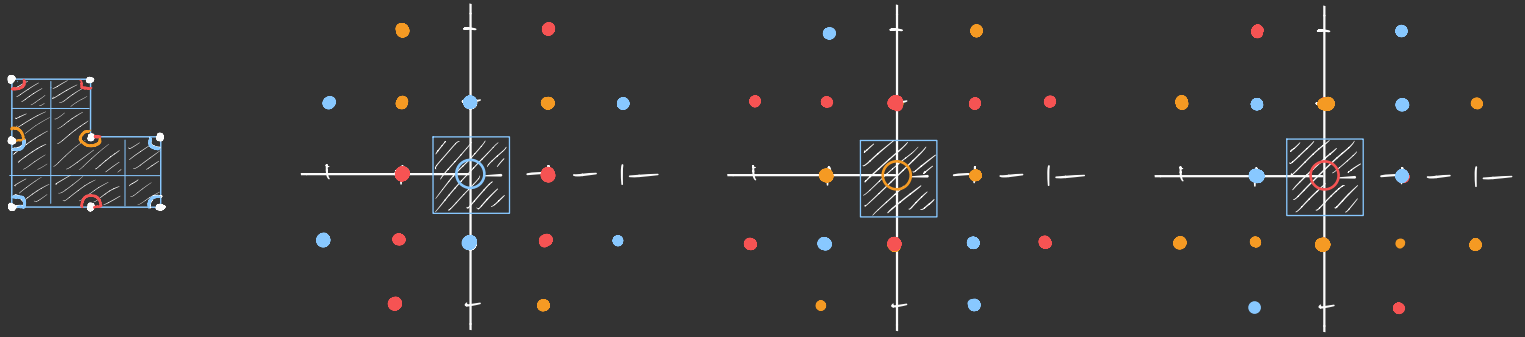
Recovering (X,w) from $\hat{\mathcal{U}}(X,w)$

- For $p \in \Theta(d_i) \subset \Theta$, let $H(p) := \{q \in \Theta(d_i) \mid d(0,q) \leq d(p,q)\}$, and for $S \subset \Theta \times \Theta$, let $\Omega(S) := \bigsqcup_{i=1}^{k} \bigcap_{\substack{\{p,p'\} \in S \\ p \in \Theta(d_i)}} H(p)$

- Defn.: The marked Voronoi staples, denoted $\hat{S}(X,w)$, are the minimal subset $S \subset \hat{\mathcal{U}}(X,w)$ for which

$$\Omega(S) = \Omega(\hat{\mathcal{U}}(X,w)).$$

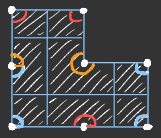
- Thm. 1 ([Ed17]): (X,w) is recovered by identifying edges of $\Omega(\hat{S}(X,w))$ via the pairing inherent to $\hat{S}(X,w)$



$SL(X,\omega)$ – membership criterion

- Thm. 2 ([Ed17], [Ess22]):

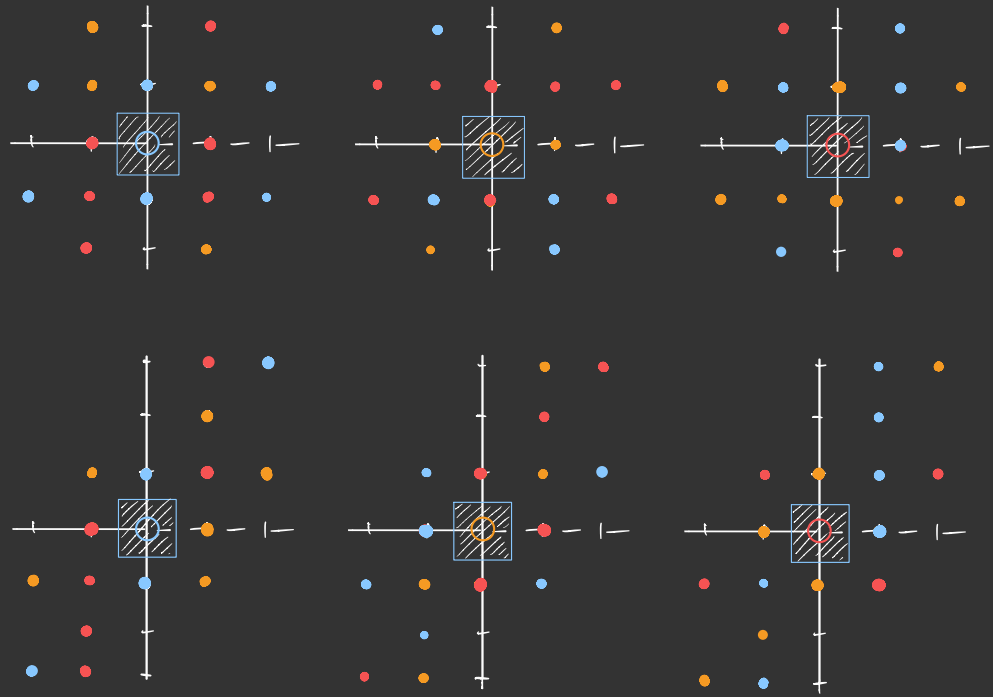
$$SL(X,\omega) = der\left(\{ f \in Aff^+(\mathbb{O}) \mid f(\vec{u}(X,\omega)) = \vec{u}(X,\omega)\}\right)$$
$$= der\left(\{ f \in Aff^+(\mathbb{O}) \mid f(\vec{s}(X,\omega)) \subset \vec{u}(X,\omega)\}\right)$$



$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

$SL(X,\omega)$

$f \in Aff^+(\mathbb{O})$, $der(f) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$



§3: Constructing lattice surfaces

History

- An implicit algorithm is given in [SW10a] to enumerate all lattice surfaces
- The current algorithm targets lattice surfaces with prescribed Veech groups

A group action on $\hat{\mathcal{M}}(X,\omega)$

- Set $Aff^+_\theta(X,\omega) := \{ f \in Aff^+(\theta) \mid f(\hat{\mathcal{M}}(X,\omega)) = \hat{\mathcal{M}}(X,\omega) \}$. Note $Aff^+_\theta(X,\omega)$ acts on $\hat{\mathcal{M}}(X,\omega)$ via

$$f(\{s,s'\}) = \{f(s), f(s')\}$$

- Recall:
  - Thm. 1: (X,ω) is determined by $\hat{S}(X,\omega) \subset \hat{\mathcal{M}}(X,\omega)$
  - Thm. 2: $SL(X,\omega) = der\left(Aff^+_\theta(X,\omega)\right)$

- Idea: 'Simulate' $Aff^+_\theta(X,\omega)$ – orbits $[\{s,s'\}] \in \hat{\mathcal{M}}(X,\omega) / Aff^+_\theta(X,\omega)$ in θ using $\Gamma = SL(X,\omega) \leq SL_2\mathbb{R}$

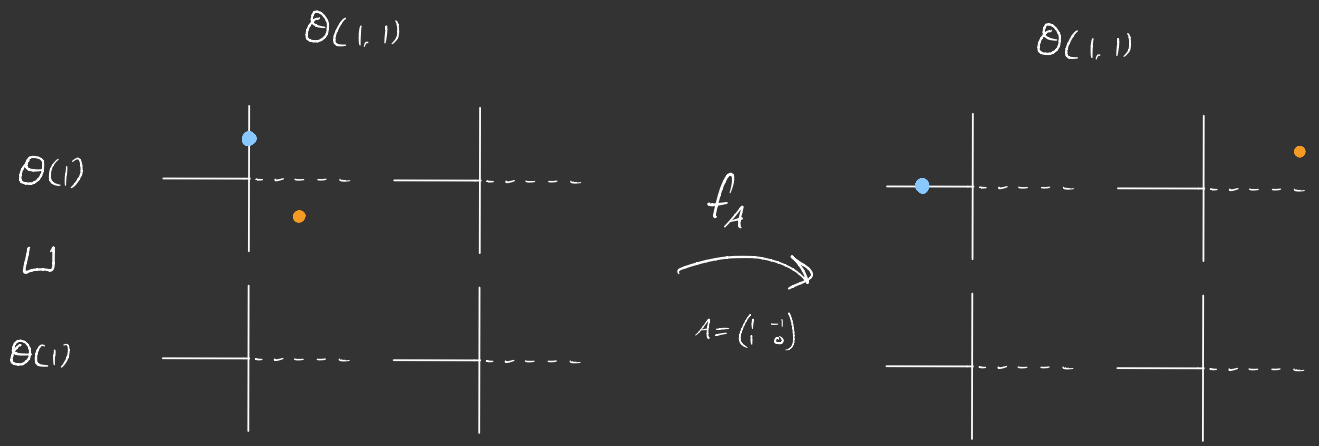
- Challenges:

  (i) Determining $Aff^+_\theta(X,\omega)$

  (ii) Determining (a) directions and (b) lengths of representatives of orbits

Canonical affine automorphisms

- Defn.: Let $Aff_c^+(\mathcal{O})$ denote the collection of $f \in Aff^+(\mathcal{O})$ satisfying $f(\mathcal{O}(d_i)) = \mathcal{O}(d_i)$ for all i, and such that the c.c. angle between $p \in \mathcal{O}$ and $f(p)$ is in $[0, 2\pi)$.
- Fact: $der(Aff_c^+(\mathcal{O})) = GL_2^+\mathbb{R}$.
- Defn.: Let f_A denote the (unique) element of $Aff_c^+(\mathcal{O})$ satisfying $der(f_A) = A$, and set $Trans(\mathcal{O}) := \ker(der)$



Determining $Aff_{\mathcal{O}}^{+}(X,\omega)$

- **Prop. 1:** Suppose $SL(X,\omega) = \Gamma = \langle A_1, \dots, A_n \rangle$. Then $Aff_{\mathcal{O}}^{+}(X,\omega)$ is generated by a subset of the finite set

$$\Gamma_{\mathcal{O}} := \{ \tau \circ f_{A_i} \mid \tau \in Trans(\mathcal{O}),\ 1 \le i \le n \}$$

- Sketch: For each i, fix $f_i := \tau_i \circ f_{A_i} \in Aff_{\mathcal{O}}^{+}(X,\omega)$, and let $\Lambda := \langle f_i \rangle_{i=1}^{n}$. It suffices to show that $Aff_{\mathcal{O}}^{+}(X,\omega) = \bigcup_{j=1}^{m} \tau_j' \Lambda$ for some $\{\tau_j'\} \subset Trans(\mathcal{O})$.

Fix $f \in Aff_{\mathcal{O}}^{+}(X,\omega)$ with $der(f) = A_{j_1}^{\delta_1} \cdots A_{j_k}^{\delta_k}$, and set $g := f_{j_1}^{\delta_1} \circ \cdots \circ f_{j_k}^{\delta_k} \in \Lambda$. Then $f \circ g^{-1} \in Trans(\mathcal{O})$, so $f = \tau \circ g$ for some $\tau \in Trans(\mathcal{O})$. Since $|Trans(\mathcal{O})| < \infty$, the result holds.

□

Determining directions of orbit representatives

- Prop. 2: For lattice $\Gamma = SL(X,\omega)$, the directions of shortest representatives of $[\{s,s'\}] \in \hat{\mathcal{U}}(X,\omega)/Aff^+_\theta(X,\omega)$ belong to the finite set

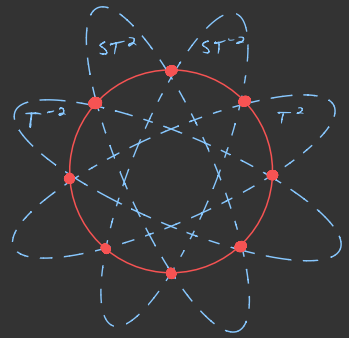
$$\Theta_\Gamma := S^1 \setminus \bigcup_{A \in \Gamma} A \cdot B_1(0)$$

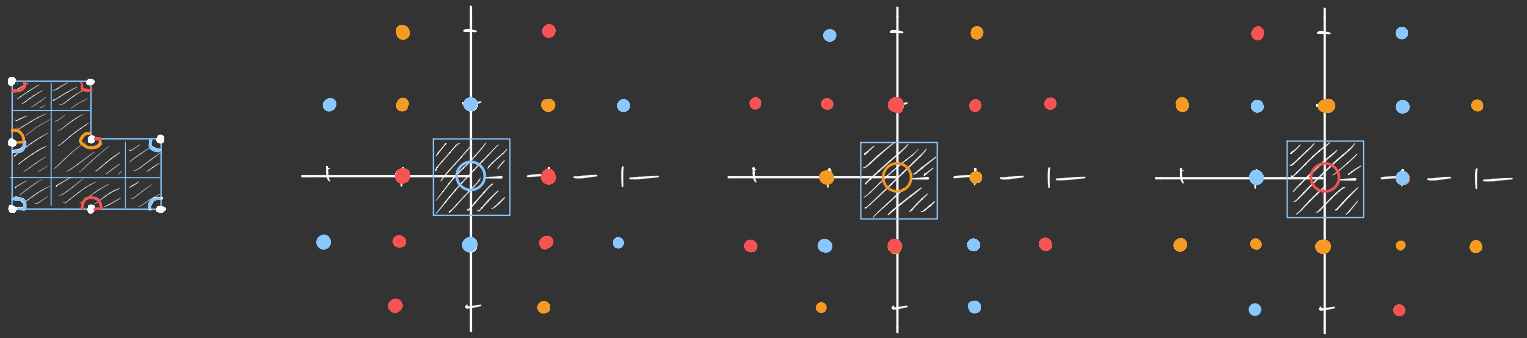
- Sketch: Let $\{s,s'\} \in [\{s,s'\}]$ be a shortest representative. Assume $|s|=1$, and suppose $\pi(s) \in A \cdot B_1(0)$ for some $A \in \Gamma$, where $\pi : \theta \to \mathbb{R}^2$ is the projection. Let $f \in Aff^+_\theta(X,\omega)$ with $der(f) = A^{-1}$. Then $|f(s)| < 1$, a contradiction.

  For lattice Γ, reciprocal slopes of elements of Θ_Γ are in bijective correspondence with points at ∞ of the 'Dirichlet domain' of Γ centered at $i \in \mathbb{H}$. Thus $|\Theta_\Gamma| < \infty$. □

Example

- Let $\Gamma = \langle S, T^2 \rangle$, where $S = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $\mathcal{H}_\Gamma \subset \{ z \in \mathbb{C} \mid z^8 = 1 \}$.





Simulating (normalized) $Aff^+_\Theta(X,\omega)$-orbits

- Let $\Gamma = \langle A_1, \dots, A_n \rangle$ be a lattice, and recall

$$T_\Theta = \{ \tau \circ f_{A_i} \mid \tau \in Trans(\Theta),\ 1 \le i \le n \} \quad \text{and} \quad \oplus_\Gamma = S^1 \setminus \bigcup_{A \in \Gamma} A \cdot B_1(0)$$

- Defn: Let $\pi : \Theta \to \mathbb{R}^2$ be the projection, and set

$$\mathcal{S} := \{ (G, (p,p')) \in \mathcal{P}(T_\Theta) \times (\pi^{-1}(\oplus_\Gamma))^2 \mid \pi(p') = -\pi(p) \}.$$

For each $s = (G, (p,p')) \in \mathcal{S}$, set $sim_0(s) := \{ \{p,p'\} \}$, and for $k \ge 1$

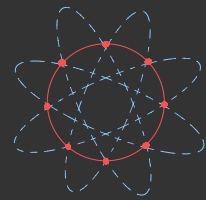
$$sim_k(s) := sim_{k-1}(s) \cup \bigcup_{\substack{f \in G \\ \{q,q'\} \in sim_{k-1}(s)}} f^{\pm}(\{q,q'\}).$$

Further, let $Sims := \{ sim(s) \}_{s \in \mathcal{S}}$, where $sim(s) := \bigcup_{k \ge 0} sim_k(s)$

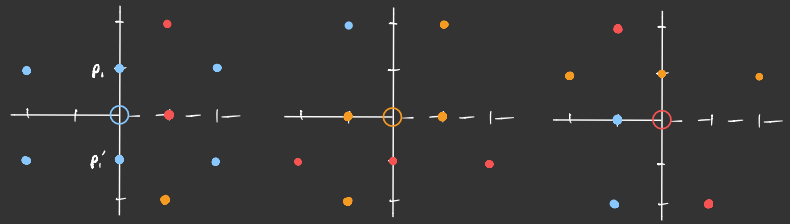
- Remark: If $(X,\omega) \in H(d_1, \dots, d_k)$ with $SL(X,\omega) = \Gamma$, then every $[\{s,s'\}]$ in $\hat{\mathcal{M}}(X,\omega) / Aff^+_\Theta(X,\omega)$ belongs to the finite set $Sims$ (up to a scalar)

Example

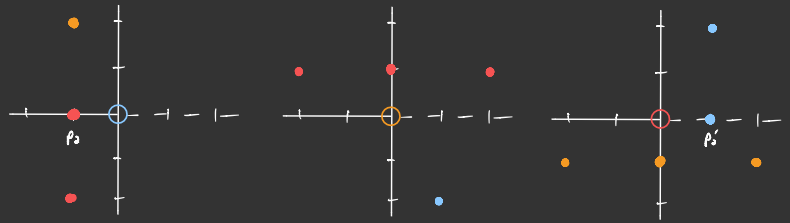
- Let $T = \langle s, T^2 \rangle$, $(d_1) = (2)$. Set $G = \{\rho^2 \circ f_s, f_T \circ \}$ with ρ a $2n$-rotation and $s_i = (G, (p_i, p_i')) \in \mathcal{S}$, $i = 1, 2, 3$



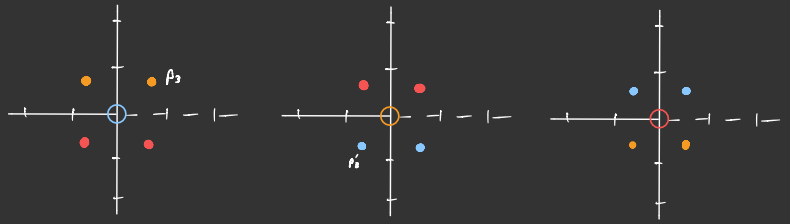
$sim(s_1) =$



$sim(s_2) =$

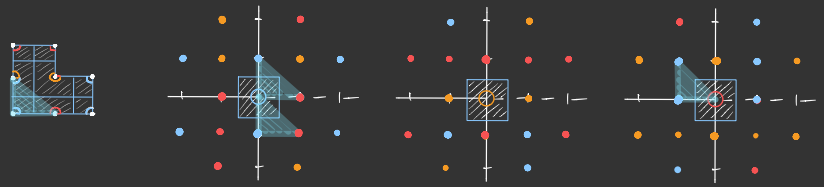


$sim(s_3) =$



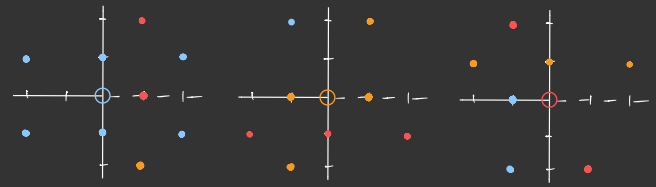
Determining lengths of orbit representatives

- Adjacent representatives of $\hat{S}(X,w)$ determine three triangles in $\mathcal{O}$

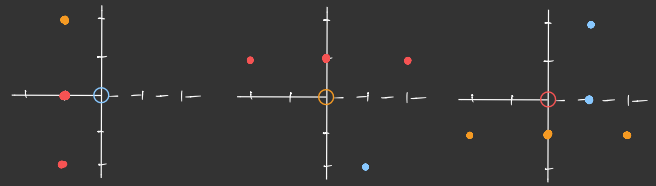


- Idea: Choose 'candidate' $\hat{S}(X,w)$ from a collection of Sims, and rescale as needed to realize the above triangles

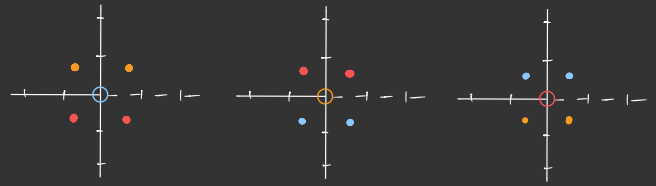
$sim(s_1)$ =



$sim(s_2)$ =



$sim(s_3)$ =



An algorithm

- Input: A lattice $\Gamma \leq SL_2\mathbb{R}$ and stratum $\mathcal{H}(d_1,\dots,d_k)$.
  Output: All $(X,\omega) \in \mathcal{H}(d_1,\dots,d_k)$ with $SL(X,\omega) = \Gamma$.

  (1) Determine $\mathcal{O}_\Gamma$ and compute $Sims_0 := \{ sim_0(s) \mid s \in \mathcal{S} \}$

  (2) For $k \geq 1$:

  a) Compute $Sims_k := \{ sim_k(s) \}_{s \in \mathcal{S}}$

  b) For $U \in \mathcal{P}(Sims_k)$ and candidate $\hat{S}(X,\omega)$ from U:

  (i) Determine scalars $\vec{r}U := \{ r_s \cdot sim_k(s) \}_{s \in \mathcal{S}}$

  (ii) Compute $\Omega(\vec{r}U)$ and construct (X,ω)

  (iii) Verify that $SL(X,\omega) = \Gamma$.

Applications

- Theorem ([SW10b]): There are at most finitely many unit-area lattice surfaces with a given Veech group in any given stratum.

- Theorem (S): The square torus is the only surface in $\bigcup_{g>0} H(2g-2)$ with Veech group $SL_2\mathbb{Z}$.

Thank you!

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